# Subject Description Form

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| **Subject Code** | ME35002 |
| **Subject Title** | Sensors and Actuators for Mechatronics |
| **Credit Value** | 3 |
| **Level** | 3 |
| **Pre-requisite/ Co-requisite/ Exclusion** | Prerequisite: Students should have basic knowledge of mechanical systems |
| **Objectives** | To provide students with the knowledge in designing mechatronic and robotic systems |
| **Intended Learning Outcomes** | Upon completion of the subject, students will be able to:<br><br>a. Formulate and solve problems related to sensors and actuators in mechatronic systems.<br>b. Understand the importance of signal conditioning and data acquisition in mechatronic systems.<br>c. Design and analyze a given task or project in a mechatronics system by applying knowledge acquired in the subject and information obtained through a literature search.<br>d. Present effectively in completing written reports of laboratory work and the given task. |
| **Subject Synopsis/ Indicative Syllabus** | Sensors - Instrumentation and measurement principles; principles of different sensors for linear and angular measurements on position, velocity, and acceleration; force and pressure sensors, principle of different cameras, 3D measurements, Time-of-flight sensors, etc.<br><br>Actuators – Different types of motors: DC, AC, servo, stepper motors; motor control shields; electric linear actuators; pneumatic systems; hydraulic systems, etc.<br><br>Signal Conditioning - Concepts and principles; complex number in electronics; operational amplifier; conversion between analog and digital signals, multiplexing; data acquisition principles, signal filtering, active and passive filters; modulation and demodulation, etc. |

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| **Teaching/Learning Methodology** | Lectures aim at providing students with an integrated knowledge required for understanding and analyzing different components (sensors and actuators) in a mechatronics system (Outcomes a to d)<br><br>Tutorials aim at enhancing students' analytical and problem-solving skills related to sensing or actuation principles. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes b to c)<br><br>The project/experiment aims to have knowledge of computer simulations and hands-on experience on mechatronics systems (Outcomes a to d) |

| **Teaching/Learning Methodology** | **Outcomes** | | | |
|---|---|---|---|---|
| | **a** | **b** | **c** | **d** |
| 1. Lectures | √ | √ | √ | √ |
| 2. Tutorials | | √ | √ | |
| 3. Project or experiments | √ | √ | √ | √ |

**Assessment Methods in Alignment with Intended Learning Outcomes**

| **Specific assessment methods/tasks** | **% weighting** | **Intended subject learning outcomes to be assessed** | | | |
|---|---|---|---|---|---|
| | | **a** | **b** | **c** | **d** |
| 1. Assignments | 10% | √ | √ | √ | √ |
| 2. Tests | 20% | √ | √ | √ | |
| 3. Projects | 20% | | √ | √ | √ |
| 4. Examinations | 50% | √ | √ | √ | √ |
| Total | 100 % | | | | |

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Overall Assessment:

**0.50 × End of Subject Examination + 0.50 × Continuous Assessment**

The examination is adopted to assess students on their overall understanding and ability to apply the concepts. It is supplemented by continuous assessment including assignments, closed-book tests, and projects. The continuous assessment is aimed at enhancing the students' comprehension and assimilation of various topics of the syllabus.

**Student Study Effort Expected**

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| Class contact: | |
| ▪ Lecture | 33Hrs. |
| ▪ Tutorial | 6Hrs. |
| Other student study effort: | |
| ▪ Reading and review | 45Hrs. |
| ▪ Coursework (assignments, project) | 36Hrs. |
| Total student study effort | 120Hrs. |

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| **Reading List and References** | 1. Bolton, W., Mechatronics: Electronic Control Systems in Mechanical Engineering, Prentice Hall, latest edition.<br>2. Shetty, D. and Kolk, R. A., Mechatronic System Design, PWS Publishing Company, latest edition.<br>3. Alciatore, D. G. and Histand, M. B., Introduction to Mechatronics and Measurement Systems, McGraw Hill, latest edition. |
| **Develop Date** | June 2024 |